

Andrew Shao

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EDUCATION

University of Michigan

Ann Arbor, MI

Bachelor of Science in Engineering: Computer Science

Aug 2022 - Dec 2025

• Relevant Coursework:

Distributed Systems, Web Systems, Operating Systems, Applied Parallel Programming with GPUs, Machine Learning, Natural Language Processing, Applied Linear Algebra

WORK EXPERIENCE

Rockwell Automation | *Software Intern*

May 2025 - Aug 2025

- Built an AI agent using LangChain, FastAPI, and Apache Kafka to analyze 2M+ CI/CD logs and detect failures, reducing manual review time by 85% and streamlining debugging via automated GitLab issue creation with suggested code fixes.
- Built an automated security tool in Python using GitLab CI/CD that integrates with the BlackDuck API to parse security vulnerability data and generate Jira tickets, streamlining the reporting workflow and improving triage efficiency by 20%.
- Developed C#/.NET 8 sample projects for the FactoryTalk Echo SDK enabling controller automation and co-simulation, while writing C++/C# unit and integration tests integrated into a Jenkins CI/CD pipeline to improve release confidence.
- Navigated a large legacy codebase for Logix Designer to resolve 5 critical bugs in C++ and C across concurrency, memory management, and legacy interface issues; utilized .NET profiling to optimize asynchronous resource management.

Sigwise | *Software Intern*

May 2023 - Aug 2023

- Developed a motion-detection Android security camera app using OpenCV, triggering video capture with intensity-based recording thresholds, and implemented a JavaMail/SMTP notification system to deliver snapshots and clips via email.
- Designed and deployed Docker containers for backend alert services, utilizing Kubernetes for cluster management and scaling, cutting environment setup time from 2 hours to under 15 minutes and streamlining testing across the team.
- Engineered a high-throughput gRPC telemetry pipeline in Go and Python to stream real-time application service metrics to a live observability dashboard, reducing monitoring collection latency from 200ms to 130ms.

University of Michigan | *Student Computer Consultant*

Oct 2022 - Mar 2026

- Built custom log-parsing Bash scripts and Zabbix monitoring dashboards to track system health and network performance across 4 campus computing sites, reducing network troubleshooting time by 30% and preventing critical service disruptions.
- Built a Chrome browser extension in Javascript that restructured helpdesk ticket data into a condensed dashboard and automated repetitive workflow actions via custom macros, greatly reducing ticket triage time for 100+ support agents.

PROJECTS

Distributed Job Scheduler

C++, gRPC, etcd/Raft

- Developed a distributed job scheduler in C++ using gRPC for inter-node communication and bin-packing to allocate jobs across 50 simulated worker nodes, sustaining 2,000 job submissions/sec with priority classes and preemption.
- Implemented heartbeat-based health checks for fault-tolerant failure detection, job rescheduling, and load balancing across nodes, backed by Raft consensus (etcd) for scheduler state with per-namespace quota enforcement and autoscaling.

Distributed Key-Value Store (Paxos & Sharded KV Store)

Go, Paxos

- Built a resilient, horizontally scalable distributed key-value store in Go using Paxos consensus and a sharded architecture across 12 shards / 30+ nodes, ensuring linearizable consistency at 15,000 ops/sec with live shard migration.
- Designed an RPC communication layer with idempotent request handling and optimized routing, achieving at-most-once semantics with p99 latency under 10ms and zero data loss across 100+ simulated node-failure trials.

MapReduce Framework & Search Engine

Python, Flask, ReactJS, Java, MapReduce, tf-idf, PageRank

- Developed a distributed MapReduce framework in Python using socket programming, multithreading, and a Manager-Worker architecture to enable fault-tolerant parallel data processing with worker monitoring and dynamic load balancing.
- Built a multi-stage data mining pipeline that processed thousands of crawled HTML documents into an inverted index, computing tf-idf, document normalization, and PageRank metrics for scalable search engine information retrieval.

SKILLS

Languages: C++/C, Go, Python, Java, C#, TypeScript, JavaScript, SQL, Bash, Kotlin, HTML/CSS

Frameworks: React, Flask, Angular, Spring, .NET, OpenCV, TensorFlow, PyTorch, Pandas/NumPy

Cloud & Tools: Google Cloud Platform, AWS, Kafka, Docker, Kubernetes, Postgres, CI/CD, CUDA, Redis, Git, Linux